
Phase 5A - SQLite Logger

Step 1: Install SQLite

Verify:

```
None
sqlite3 --version
```

If not installed:

```
None
sudo apt install sqlite3
```

Step 2: Install Python MQTT package

```
None
pip3 install paho-mqtt
```

Step 3: Create logger file

```
None
nano vehicle_logger.py
```

Paste:

```
None
import sqlite3
import paho.mqtt.client as mqtt
import re
```

```
# ----------------------------
# SQLite Setup
# ----------------------------

conn = sqlite3.connect("vehicle_telematics.db")
cursor = conn.cursor()

cursor.execute("""
CREATE TABLE IF NOT EXISTS location(
   id INTEGER PRIMARY KEY AUTOINCREMENT,
   timestamp DATETIME DEFAULT CURRENT_TIMESTAMP,
   latitude REAL,
   longitude REAL
)
""")

cursor.execute("""
CREATE TABLE IF NOT EXISTS status(
   id INTEGER PRIMARY KEY AUTOINCREMENT,
   timestamp DATETIME DEFAULT CURRENT_TIMESTAMP,
   speed INTEGER,
   fuel INTEGER
)
""")

cursor.execute("""
CREATE TABLE IF NOT EXISTS diagnostics(
   id INTEGER PRIMARY KEY AUTOINCREMENT,
   timestamp DATETIME DEFAULT CURRENT_TIMESTAMP,
   rpm INTEGER,
   engine_temp INTEGER
)
""")

conn.commit()
```

```
# ----------------------------
# MQTT Callback
# ----------------------------

def on_message(client, userdata, msg):

   payload = msg.payload.decode()

   print(f"\nTopic: {msg.topic}")
   print(f"Payload: {payload}")

   if msg.topic == "vehicle/location":

       lat = float(
           re.search(
               r'Latitude=([\d.]+)',
               payload).group(1)
       )

       lon = float(
           re.search(
               r'Longitude=([\d.]+)',
               payload).group(1)
       )

       cursor.execute(
           "INSERT INTO location(latitude,longitude) VALUES(?,?)",
           (lat, lon)
       )

   elif msg.topic == "vehicle/status":

       speed = int(
           re.search(
               r'Speed=(\d+)',
```

```
                payload).group(1)
        )

        fuel = int(
            re.search(
                r'Fuel=(\d+)',
                payload).group(1)
        )

        cursor.execute(
            "INSERT INTO status(speed,fuel) VALUES(?,?)",
            (speed, fuel)
        )

    elif msg.topic == "vehicle/diagnostics":

        rpm = int(
            re.search(
                r'RPM=(\d+)',
                payload).group(1)
        )

        temp = int(
            re.search(
                r'EngineTemp=(\d+)',
                payload).group(1)
        )

        cursor.execute(
            "INSERT INTO diagnostics(rpm,engine_temp) VALUES(?,?)",
            (rpm, temp)
        )

    conn.commit()

# ----------------------------
```

```
# MQTT Setup
# ----------------------------

client = mqtt.Client()

client.on_message = on_message

client.connect("localhost", 1883, 60)

client.subscribe("vehicle/location")
client.subscribe("vehicle/status")
client.subscribe("vehicle/diagnostics")

print("Vehicle Logger Started...")

client.loop_forever()
```

Save:

```
None
CTRL + O
ENTER
CTRL + X
```

Step 4: Run Logger

Terminal 1:

```
None
python3 vehicle_logger.py
```

```
aniruddhan@raspberrypi:~ $ nano vehicle_logger.py
aniruddhan@raspberrypi:~ $ python3 vehicle_logger.py
/home/aniruddhan/vehicle_logger.py:115: DeprecationWarning: Callback API version 1 is deprecated, update to latest version
  client = mqtt.Client()
Vehicle Logger Started...

Topic: vehicle/location
Payload: Latitude=13.0949 Longitude=80.2829

Topic: vehicle/status
Payload: Speed=59 Fuel=11

Topic: vehicle/diagnostics
Payload: RPM=692 EngineTemp=82

Topic: vehicle/location
Payload: Latitude=13.0951 Longitude=80.2831

Topic: vehicle/status
Payload: Speed=35 Fuel=13

Topic: vehicle/diagnostics
Payload: RPM=1896 EngineTemp=82

Topic: vehicle/location
Payload: Latitude=13.0953 Longitude=80.2833
```

Step 5: Run Publisher

Terminal 2:

```
None
./tcu_mqttCAN
```

Step 6: Verify Database

Open database **in a separate Terminal**:

```
None
sqlite3 vehicle_telematics.db
```

Show tables:

```
None
.tables
```

Expected:

```
None
diagnostics
location
status
```

Check records:

```
None
SELECT * FROM location LIMIT 5;
```

```
None
SELECT * FROM status LIMIT 5;
```

```
None
SELECT * FROM diagnostics LIMIT 5;
```